

Songhang Li

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EDUCATION

Columbia University <i>M.S. in Electrical Engineering</i> <i>B.S. in Electrical Engineering</i>	New York, NY May 2028 May 2027
Franklin & Marshall College <i>B.A. in Physics and Mathematics</i> <i>Magna Cum Laude</i> Φ BK	Lancaster, PA May 2025

AWARDS & HONORS

Frank Durrell Enck Memorial Prize in Physics <i>F&M</i> Established in 1975; traditionally awarded to a graduating Physics major whose overall departmental performance and involvement is most deserving of recognition.	Spring 2025
Kershner Scholar in Mathematics <i>F&M</i> Awarded to students based on their proficiency in Mathematics.	Spring 2025
Phi Beta Kappa (ΦBK) <i>U.S.A.</i> Invitation-only academic honor society recognizing excellence in the liberal arts and sciences.	Since 2025
The Nolt Music Award <i>F&M</i> Awarded to F&M student musicians to undertake musically enriching projects such as a senior recital.	Spring 2025
The Alma Mater Singer <i>F&M (238th Commencement)</i> Selected as the Alma Mater Singer for Franklin & Marshall College's 238th Commencement.	Spring 2025
Summer Scholar Award <i>F&M</i> Awarded to a student researcher for summer research conducted with faculty.	Summer 2024
Pi Mu Epsilon (Math Honor Society) <i>U.S.A.</i> Nationally recognized honorary society dedicated to furthering knowledge of mathematics.	Since Spring 2024
Sigma Pi Sigma (Physics & Astronomy Honor Society) <i>U.S.A.</i> Honors outstanding scholarship in physics and astronomy.	Since Spring 2023
Student Independent Research Grants <i>F&M</i> Awarded for advanced student research/projects whose needs exceed departmental budgets.	Fall 2023
Kershner Scholar in Physics <i>F&M</i> Awarded to students based on their proficiency in Physics.	Spring 2023
Dean's and Honors Lists <i>F&M</i> Earned Honors List recognition for achieving a 3.7 or better grade point average.	Since Fall 2021

PUBLICATIONS

J.K. Krebs, **Songhang Li**, Dragos Georgescu, and T. D. Pham. (2026). *Eu3+ Luminescence as a Probe of Local Crystallinity in Combustion-Synthesized LaAlO₃*. *Materials Letters*, 414. [doi:10.1016/j.matlet.2026.140581](https://doi.org/10.1016/j.matlet.2026.140581)

Eli Johnson, Max Drucker, Manuel Garcia, **Songhang Li** (2026). *A Programmable Four-Topology Amplifier Chip for Analog Education*. *MIT URTC* | *IEEE Xplore* in prep.

Yuxuan Du, Wenya Du, Dabin Lin, Minghao Ai, **Songhang Li**, Lin Zhang. (2023). *Recent Progress on Hydrogel-Based Piezoelectric Devices for Biomedical Applications*. *Micromachines*, 14(1). [doi:10.3390/mi14010167](https://doi.org/10.3390/mi14010167)

PRESENTATIONS & CONFERENCES

- MIT Undergraduate Research Technology Conference, Mid-Atlantic Section** | *MIT, MA* Oct. 9, 2026
- A Programmable Four-Topology Amplifier Chip for Analog Education.
- American Physical Society Meeting, Mid-Atlantic Section** | *Temple University, PA* Nov. 15, 2024
- Rate equation model for fluorescence and phosphorescence in Eu^{3+} doped LaAlO_3 .
- Student Research Fair** | *Franklin and Marshall College, PA* Oct. 18, 2024
- Photon Counting Experiment of Persistent Luminescence in Eu:LAP
- Student Research Fair** | *Franklin and Marshall College, PA* Apr. 20, 2024
- Design, construction, testing, and improving: An introduction to Engineering Design Competitions.

ACADEMIC & RESEARCH EXPERIENCES

- Laser Induced Transfer Forwarding vs E-beam evaporation on PVDF** May 2026 – Present
Undergraduate Researcher Advisor: Prof. Ioannis Kyriassis, Peter Ballentine *Columbia University, NY*
- Investigating LIFT as a low-thermal-budget to E-beam evaporation for PVDF electrode deposition, evaluating its ability to preserve piezoelectric performance(β -phase) through comparative characterization.
 - Developing a data acquisition system integrating an oscilloscope and ADMET EP1 electromechanical controller to quantify the relationship between applied force rate and PVDF charge output.
 - Designing electrode and interconnect layouts in KLayout and preparing vinyl stencil.
 - Fabricating screen-printed silver-paste electrodes, enabling low-temperature processing to preserve PVDF piezoelectric properties.
- IEEE SSCS 2026 Chipathon Integrated Circuit Design Contest** May 2026 – Present
Analog IC Designer Advisor: Prof. Peter Kinget, Luigi Viton-Zorrilla *Columbia University, NY*
- Designing a programmable five-transistor operational transconductance amplifier for the open-source Four-Topology MOSbius test chip in the GF180MCU 180 nm process.
 - Applying gm/Id-based transistor sizing and Xschem/ngspice simulations to evaluate gain, bandwidth, stability, slew rate, power consumption, and output range across programmable sizing modes.
 - Completing the OTA layout in KLayout with DRC and LVS clean verification.
 - Preparing tapeout by simulating inverting op amp implementation for top-level verification.
 - Preparing a manuscript for submission in *MIT Undergraduate Research Technology Conference* and *IEEE Xplore*.
- Photon Counting Experiment of Persistent Luminescence in Eu:LAP** May 2024 – Jan. 2025
Research Assistant Advisor: Prof. Ken Krebs *F&M, Lancaster, PA*
- Measured decay dynamics of excited population of impurity ions after UV laser excitation using photon counting.
 - Developed a circuit solution using Arduino to precisely control excitation laser mechanical shutter, enabling photon counting of early state decay.
 - Compared the emission spectra measurement with calculated energy levels of Eu^{3+} in LaAlO_3 using density functional theory.
 - Conducted a literature review and originated a table summarizing experimental technique and materials.
 - Prepared manuscript, see publication (Materials Letters, 414.)
- Society of Automotive Engineers Aero Design Competition Preparation** Sept. 2023 – May 2025
Team Lead, Pilot Advisor: Prof. Ken Krebs *F&M, Lancaster, PA*
- Designed a power and lift efficient, short-take-off-length 3.8 meters fixed wing plane for water delivery.
 - Calculated the lift of the wing based on the structure of air foil in order to improve the load capacity.
 - Fabricated the wings utilizing balsa wood and carbon fiber, employing laser cutter and heat-shrink covering film application techniques.
 - Led the team members to write the design report and presented the result in a research fair.
 - Wrote a funding proposal and managed to receive a year-long student research grant from the grant committee.
- Electrical and Optical Measurement of Ferroelectrics Material (PLZT)** Sept. 2022 – Dec. 2022
Researcher Advisor: Prof. Ken Krebs *F&M, Lancaster, PA*
- Investigated the relation between the refractive index and dielectric constants of PLZT film.
 - Built multilayer models by ellipsometry data analysis software CompleteEASE to model measurements of the thickness of thin films.

- Measured the refractive index using J.A. Woollam M-2000 spectrum ellipsometry.
- Made a two-plate capacitor with PLZT as dielectric layer to obtain the dielectric constant from capacitance measurements.
- Compared optical and electrical techniques for determination of dielectric constant and refractive index.

Universal Village Program-Smart Materials and Intelligent Electronics Jun. 2022 – Aug. 2022

Research Assistant Advisor: Dr. Lin Zhang, Massachusetts Institute of Technology Media Lab Boston, MA

- Designed a flexible electrode pattern exposure mask using AutoCAD, in preparation for printing of mask.
- Prepared piezoelectric 1-3 composites for piezoelectric sensors.
- Organized the piezoelectric hydrogel sensors comparison table and contributed to manuscript writing, see publication (Micromachines 2023, 14, 167).

SKILLS & INSTRUMENTATION

Hardware & Instrumentation: ADMET Testing System, Oscilloscopes, Waveform Generators, HP LCR, M-2000 J.A. Woollam Spectrum ellipsometry, Monochromator SR430, Stanford Research System Multichannel Photon Counter.

EDA Tools & Design Software: LT SPICE, NG SPICE, MATLAB, Xschem, KLayout, Arduino IDE, Onshape, Cadence.

Programming language: Python, Java, MATLAB, MIPS Assembly, Mathematica, R.

Software: IntelliJ, AutoCAD, MATLAB, CompleteEASE(optical modeling and ellipsometry data fitting)

Design & Analysis Skills: Electromechanical characterization, Analog and digital circuit simulation (AC/DC/transient analysis), Circuit design and debugging, Data analysis with Python/MATLAB/SciDavis, Characterization of electronic components and semiconductor materials.

Additional Technical Skills: Laser cutting/fabrication (Glowforge), Vinyl cutter, 3D printing, PCB prototype including exposure and etching technics, technical documentation (LaTeX, Endnote), design visualization and project planning (X-mind), image processing (Adobe Lightroom, Photoshop), Screen printing.

EXPERIENCE

Department of Electrical Engineering Feb. 2026 – May 2026

Course Assistant Columbia University, New York, NY

- Prepared the LED matrix via passive scanning using Arduino for ENGI E1102: *The Art of Engineering*.

Department of Physics and Astronomy Jan. 2024 – May 2025

Teaching and Research Assistant Franklin & Marshall College, Lancaster, PA

- Compared two electronics textbooks and exercises to identify key instructional concepts for PHY321 Intro to Electronics.
- Tested different teaching kits selected by the professor and evaluated suitability for effective teaching and learning.

Career Center Sept. 2022 – May 2025

Photographer Franklin & Marshall College, Lancaster, PA

- Captured LinkedIn headshots and event photography for Career Services programming.

REFERENCES

Ioannis (John) Kymissis Professor of Electrical Engineering, Columbia University

Charles Zukowski Professor of Electrical Engineering, Former EE Department Chair, Columbia University

Ken Krebs Professor of Physics, Franklin & Marshall College

Lin Zhang Assistant Professor of Biomedical Engineering, University at Buffalo The State University of New York

Calvin Stubbins Professor of Physics, Franklin & Marshall College

Micheal McCooey Associate Professor of Mathematics, Franklin & Marshall College

Savannah Eisner Assistant Professor of Electrical Engineering, Columbia University

COURSEWORK

Electrical Engineering: Electronic Circuit, Solid State Devices & Material, Signal & System, Communication System, Modern Display Science & Technology.

Physics: Advanced Methods in Physics, Electromagnetism, Quantum Mechanics, Analytic Mechanics, Mathematic Methods in Physics, Statistical & Thermal Physics.

Math: Advanced Linear Algebra, Abstract Algebra, Multi-variable Calculus, Probability & Statistics, Differential Equation, Origami Math, Real Analysis.

ACTIVITIES

Vocal Sept. 2021 – May 2025
Recital & College Chorus Singer *F&M*

- Sang the Alma Mater at the Class of 2025 Commencement.
- Performed with the College Chorus at major campus events (e.g., True Blue Weekend).
- Performed in end-of-semester vocal lesson recitals each semester.

Alumni Leaders of Tomorrow, Office of Alumni Engagement Jun. 2023 – May 2024
Member *F&M*

- Created alumni connection and networking programs to support student–alumni engagement.

F&M Tennis Club Sept. 2023 – May 2025
Member *F&M*

- Trained and onboarded new club members in collaboration with the coach.

INTEREST

RC airplane, singing, photography (fashion and documentary, film, and darkroom processing), tennis, swimming, cycling.